

Derivation of the Kerr spin-suppression factor

Kerr Hawking
temperature
 $T_K = \frac{\hbar c(r_+ - r_-)}{4\pi k_B(r_+^2 + a^2)}$

D3 formula
[Sharma 2026]
 $\Delta r = 2Gk_B T \ln 2/c^4$

Ratio to
Schwarzschild
 $f(\chi) = T_K/T_S$

Key step: $s^2 + \chi^2 = 1$ collapses $(1+s)^2 + \chi^2 \rightarrow 2(1+s)$

EXACT CLOSED FORM
[this work]

$$f(\chi) = \frac{2\sqrt{1-\chi^2}}{(1+\sqrt{1-\chi^2})}$$